SCMission Contest 2026

Round 2.2: FRONTIER SHIFT

TEAM: SOTA


Bùi Minh Phương


Trần Thu Hà


Phạm Đức Trí


Tân Hồng Phong

Executive summary

Situation

The Company: MK Learn & Play is a premium Vietnamese toy manufacturer exporting to the US, UK, and Australia .



Current State: Goods have been produced in the Vietnam factory and are ready for global distribution.



Complication

- **Logistics (Part A)**: Balancing distribution plan, lead times, total landed costs, and container capacities across four distinct DCs .
- **Partners (Part B)**: Vulnerability to local distributor failures and the lack of fallback options .
- **Disruptions (Part C)**: A sudden port congestion event adds 15 days to ocean lead times and spikes costs by 20%, threatening high-priority orders.

Question

How can MK design an end-to-end distribution pipeline and leverage technology to ensure a 98% fulfillment rate for its global customers?

Answer

- **Part A (Logistics Plan)**: Design a pipeline to allocate shipments optimally across markets, justifying shipping modes to minimize total landed cost .
- **Part B (Distributor Selection)**: Select optimal local partners based on capacity and consumer reach, while establishing clear mitigation plans for fulfillment drops .
- **Part C (Intelligence System)**: Deploy an AI-powered Distribution Intelligence System (DIS) to monitor KPIs, manage fragmented shipments between DC-US-W and DC-US-E, and automatically recommend cost-effective routing changes (e.g., airfreight vs. ocean) during sudden disruptions.

Assumption

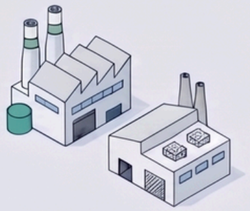
Production pulled forward by 1 week: Round 2.1 production is assumed to finish one week earlier.	**Ship immediately after production:** Finished goods are shipped once completed; the previous 1-week waiting gap is removed.
Peak windows as readiness benchmark: ETA is evaluated against the relevant peak demand windows.	**US East and West as separate lanes:** DC-US-E and DC-US-W loads cannot be consolidated in the same container.
Part A keeps planned destinations: Baseline shipments follow their original assigned DCs.	**Part C has one destination exception:** Only Product A shipment of **13,663 units** is re-evaluated/redirected for the UK case.
Split shipment = 40/60: 40% urgent volume by air, 60% remaining volume by sea.	**Planner-approved DIS:** DIS recommends actions; planners approve final execution.

End-to-End Supply Chain Snapshot

KEY METRICS

Factory

- **Demand**: **680,312** units
- **Shortage**: **88,174** units / **13.0%**
- **Main bottleneck: 456CD**
- **456CD shortage: 44,965** units / **51.0%** of total shortage



Factory is the core supply constraint. Downstream can only optimize what the factory can produce.

Logistics

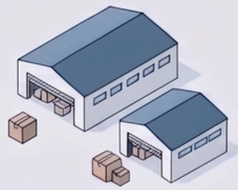
- Base ETA vs New ETA: **+15 days** on affected lanes
- Delayed: **38 shipment rows**
- Affected DC / lane**: DC-US-E and DC-UK**
- Split saving vs full air: **$720,677**
- **Schedule Impact:** 7 shipments change to SS2



0 RED does not mean 0 risk: 7 schedule misses (SS1 to SS2) confirm that ETA delays are significantly thinning DC buffers.

Distribution Center

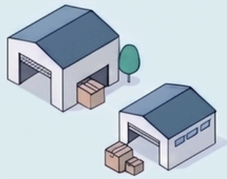
- DCs below buffer: **2 / 4**
- Weak DCs: **DC-US-E, DC-UK**
- Safety threshold: **1 week**
- DC-US-E and DC-UK: **below 1-week coverage**
- **Storage capacity**



DCs absorb logistics shocks. Weak buffer can turn a 15-day delay into stockout risk.

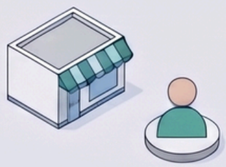
Distributor

- Distributor SLA risk: **1 / 3**
- Channel fit
- Planned CBM vs capacity
- PlayGiant USA rolling SLA: **92.75%**
- SLA threshold: **95%**
- Backup: **KidWorld Distrib.**



US risk also depends on distributor SLA. If PlayGiant stays below 95%, backup must be ready.

Customer



Customer service is a chain outcome. It depends on factory supply, ETA reliability, DC buffer, and distributor SLA.

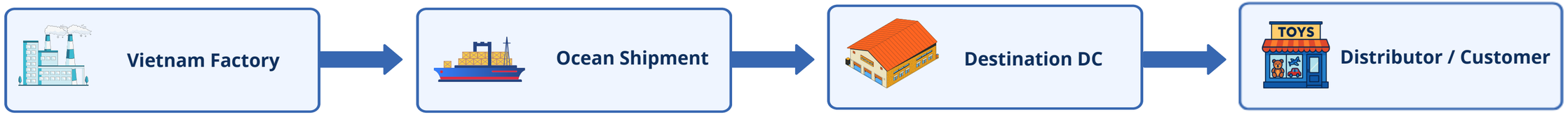
Top recommendation	Key evidence	What MK should do	Expected impact
1. Prioritize constrained factory supply before logistics execution	Demand: **680,312 units**; shortage: **88,174 units / 13.0%**; main bottleneck: **Fixture 456CD**	Protect high-priority SKUs/orders first; allocate bottleneck capacity by priority; release priority flag to logistics	Prevents downstream from optimizing unavailable supply; protects the most critical demand first
2. Use targeted split shipment, not full-network airfreight	**38 delayed shipments**; max delay **15 days**; readiness **54G / 10A / 0R**; full air **$1.276M** vs split **$555.6K**	Keep standard volume on sea; use split shipment for tracked/high-risk loads; reserve full air only for critical cases	Protects urgent demand while avoiding excessive airfreight cost; saves **$720.7K vs full air**
3. Operate DIS as a weekly exception-control tower	**2/4 DCs** below buffer: DC-US-E, DC-UK; PlayGiant USA SLA **92.75%** vs 95% threshold	Monitor ETA delay, DC coverage, and distributor SLA together; trigger expedite, rebalance, or backup action	Detects service risk before customer failure; links shipment delay → DC buffer → distributor SLA

A1. The Distribution Pipeline Timeline Planning

The pipeline is planned at load level, allowing every positive SKU-load line to be traced from Vietnam departure to destination DC arrival across W04–W15.

Planning Window	Shipment Lines	Unique Loads	Total Units	Total CBM
W04–W15	64	37	236,356	626.76

End-to-end pipeline logic



Destination DC summary

Destination DC	Market	Units	CBM	Loads
DC-US-W	US West	110,996	233.28	11
DC-US-E	US East	73,996	145.94	8
DC-UK	UK	49,164	240.66	12
DC-AU	Australia	2,200	6.88	6

Planning insights

- **US is the execution priority: 184,992 units** are split across two DCs, so the pipeline must control DC assignment at load/container level.
- **UK is CBM-heavy:** although UK has fewer units than US, it carries **240.66 CBM,** slightly higher than each US DC, making it important for freight and warehouse capacity planning.
- **AU remains ultra-lean:** only **2,200 units / 6.88 CBM,** so AU should stay flexible and is more suitable for LCL-based replenishment.

Bottom line: A1 converts the shipment plan into a traceable factory-to-DC pipeline, showing which loads go to which DCs, when they arrive, and where execution risks may emerge

Excel reference: Pipeline Master Table (row-level logic) + All_Load_Detail (source load data).

A1. The Distribution Pipeline Timeline Planning

DC allocation is first defined by market, UK and AU are assigned to single-market DCs, while the US is split between DC-US-W and DC-US-E using a weighted scoring matrix and whole-container assignment.

US DC Allocation Scoring Matrix - DC-US-W vs DC-US-E			
Criteria	**Weight**	**DC-US-W Score**	**DC-US-E Score**
Storage Capacity Fit	20%	10	8
DC Handling Cost Efficiency	15%	10	9
Baseline Lead-time Compliance	10%	10	10
Disruption Resilience	35%	10	3
US Customer Channel & Peak Demand Fit	20%	9	7.5
Calculated Weighted Total	100%	**9.8**	**6.5**
Score-based Allocation Ratio		**60.12%**	**39.88%**
Rounded Operating Allocation		**60%**	**40%**

Market	Destination DC logic	Allocation rule
US	Split between DC-US-W and DC-US-E	**60/40 by whole containers**
UK	Assigned to DC-UK	**Single-market DC**
AU	Assigned to DC-AU	**Single-market DC**



DC-US-W 60% DC-US-E 40%

Actual US allocation after whole-container assignment					
Destination DC	**Role in plan**	**Units**	**Actual ratio**	**Target ratio**	**Execution logic**
DC-US-W	**Primary / lower cost & resilient**	110,996	**60.00%**	60%	**Closest feasible split by whole containers**
DC-US-E	**Secondary / US East coverage**	73,996	**40.00%**	40%	**Kept as DC-level loads; no SKU splitting**

Weighting logic reflects distribution risk priority:

- **Disruption resilience** receives the highest weight **(35%)** because DC allocation must protect service continuity under port, capacity, or lead-time disruption.
- **Storage capacity fit and peak-demand fit** are weighted **20%** each to ensure the selected DC can absorb seasonal volume without creating downstream bottlenecks.
- **Handling cost efficiency** is weighted **15%** because cost matters but cannot override service reliability during peak distribution.
- **Baseline lead-time compliance** is weighted **10%** because both US DCs meet the required standard; therefore, it serves as a qualifying condition, not the main decision driver.

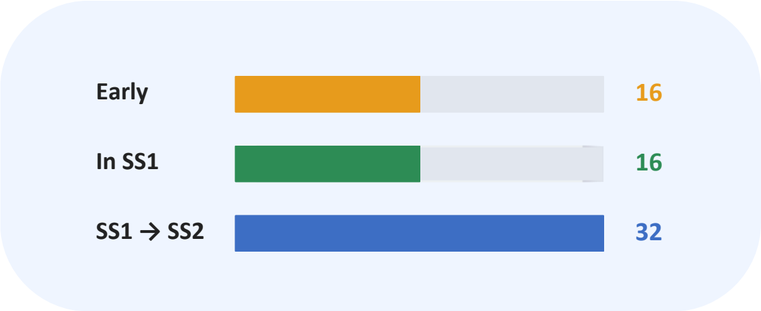
Why this matters: The DC split is operationally executable because allocation is locked at container/load level, preventing SKU-level fragmentation within the same shipment

Excel reference: Note for A.1 (US scoring matrix + allocation audit) and Pipeline Master Table (Destination DC by row).

A1. The Distribution Pipeline Timeline Planning

Each shipment line is checked against its destination DC lead time and seasonal demand window, turning departure and arrival dates into clear execution actions.

Status	Lines	Share	Proposed action	Operational meaning
Early before SS1	16	25%	Hold at DC / consolidate until SS1	Creates controllable inventory buffer
In SS1 window	16	25%	Ship as planned / prioritize SS1	Directly supports SS1 demand
Between SS1 & SS2	32	50%	Hold or reallocate for SS2 demand	Pre-positions stock for the next wave
Total	64	100%	Every line has an action	No baseline line is left undefined



Market	Window	Season name	Window start
US	SS1	Back-to-school	**1/8/2026**
US	SS2	Christmas	**1/10/2026**
UK	SS1	Easter	**1/3/2026**
UK	SS2	Christmas	**1/11/2026**
AU	SS1	July School Holiday	**1/7/2026**
AU	SS2	Christmas	**1/11/2026**

Planning insights

- **All 64 shipment lines are action-ready: each line has a destination DC, expected arrival date, timing status, and proposed action.**
- **Early arrivals are not treated as problems: they become controllable DC inventory buffers before the relevant market window.**
- **In-window arrivals should be prioritized: these shipments can directly support the first applicable seasonal demand period.**
- **Post-window arrivals still create value: shipments arriving after the first window are retained or redirected to support the next seasonal replenishment cycle.**

Bottom line: A1 converts the production output into an auditable distribution timeline, showing whether each shipment arrives early enough for its market window and what MK should do next

Excel reference: Pipeline Master Table (Seasonal Status, Proposed Action, Destination DC, Load Type, CBM columns).

A2. Shipping mode decision & Cost justification

Shipping mode to US

Baseline Uses a hybrid FCL+LCL strategy to ensure 100% container utilization and minimize landed cost.

Market	Week	Week_Date	Units_A	Units_B	Units_F	Total_Demand_CBM	n40	n20	LCL_CBM	Coverage_CBM	Slack_CBM	Cost
US	4	2026-05-11	15000,00	0,00	3000,00	40,4297	1	0	0,0000	65,0000	24,5703	$5.200,00
US	5	2026-05-18	12500,00	0,00	2500,00	33,6953	0	1	5,6953	33,6953	0,0000	$4.139,06
US	6	2026-05-25	11000,00	4000,00	3000,00	36,6797	0	1	8,6800	36,6800	0,0003	$4.736,00
US	7	2026-06-01	13000,00	2000,00	2700,00	38,2656	0	1	10,2656	38,2656	0,0000	$5.053,12
US	8	2026-06-08	13773,00	1227,00	3000,00	39,2884	1	0	0,0000	65,0000	25,7116	$5.200,00
US	9	2026-06-15	7058,00	5309,00	3000,00	28,8797	0	1	0,8813	28,8813	0,0016	$3.176,25
US	10	2026-06-22	2040,00	5309,00	2700,00	16,0406	0	1	0,0000	28,0000	11,9594	$3.000,00
US	11	2026-06-29	3541,00	3081,00	3000,00	16,6084	0	1	0,0000	28,0000	11,3916	$3.000,00
US	12	2026-07-06	12041,00	2959,00	2300,00	36,9850	0	1	8,9850	36,9850	0,0000	$4.797,00
US	13	2026-07-13	12040,00	2960,00	2700,00	37,3656	0	1	9,3656	37,3656	0,0000	$4.873,12
US	14	2026-07-20	9246,00	3000,00	2700,00	30,4481	0	1	2,4481	30,4481	0,0000	$3.489,62
US	15	2026-07-27	5754,00	4800,00	2700,00	24,5306	0	1	0,0000	28,0000	3,4694	$3.000,00

Alterrnative Overbooks FCLs to avoid LCL delays, wasting money on shipping "empty air" (+280% Slack).

Market	Week	Week_Date	Units_A	Units_B	Units_F	Total_Demand_CBM	n40	n20	LCL_CBM	Coverage_CBM	Slack_CBM	Cost
US	4	2026-05-11	15000,00	0,00	3000,00	40,4297	1	0	0,0000	65,0000	24,5703	$5.200,00
US	5	2026-05-18	12500,00	0,00	2500,00	33,6953	0	1	5,6953	33,6953	0,0000	$4.139,06
US	6	2026-05-25	11000,00	4000,00	3000,00	36,6797	1	0	0,0000	65,0000	28,3203	$5.200,00
US	7	2026-06-01	13000,00	2000,00	2700,00	38,2656	1	0	0,0000	65,0000	26,7344	$5.200,00
US	8	2026-06-08	13773,00	1227,00	3000,00	39,2884	1	0	0,0000	65,0000	25,7116	$5.200,00
US	9	2026-06-15	7058,00	5309,00	3000,00	28,8797	0	1	0,8800	28,8800	0,0003	$3.176,00
US	10	2026-06-22	2040,00	5309,00	2700,00	16,0406	0	1	0,0000	28,0000	11,9594	$3.000,00
US	11	2026-06-29	3541,00	3081,00	3000,00	16,6084	0	1	0,0000	28,0000	11,3916	$3.000,00
US	12	2026-07-06	12041,00	2959,00	2300,00	36,9850	1	0	0,0000	65,0000	28,0150	$5.200,00
US	13	2026-07-13	12040,00	2960,00	2700,00	37,3656	1	0	0,0000	65,0000	27,6344	$5.200,00
US	14	2026-07-20	9246,00	3000,00	2700,00	30,4481	0	1	2,4481	30,4481	0,0000	$3.489,62
US	15	2026-07-27	5754,00	4800,00	2700,00	24,5306	0	1	0,0000	28,0000	3,4694	$3.000,00

Shipping mode to UK

Baseline Shifts residual off-peak volume to LCL, trading a 5-day delay for maximum cost efficiency.

Market	Week	Week_Date	Total_Demand_CBM	n40	n20	LCL_CBM	Coverage_CBM	Slack_CBM	Cost	Units_C	Units_D
UK	4	2026-05-11	24,9900	0	0	24,9900	24,9900	0,0000	$1.749,30	4998,00	0,00
UK	5	2026-05-18	20,8300	0	0	20,8300	20,8300	0,0000	$1.458,10	4165,00	0,00
UK	6	2026-05-25	25,0000	0	0	25,0000	25,0000	0,0000	$1.750,00	5000,00	0,00
UK	7	2026-06-01	25,0000	0	0	25,0000	25,0000	0,0000	$1.750,00	5000,00	0,00
UK	8	2026-06-08	25,0000	0	0	25,0000	25,0000	0,0000	$1.750,00	5000,00	0,00
UK	9	2026-06-15	25,0000	0	0	25,0000	25,0000	0,0000	$1.750,00	5000,00	0,00
UK	10	2026-06-22	25,0000	0	0	25,0000	25,0000	0,0000	$1.750,00	5000,00	0,00
UK	11	2026-06-29	25,0000	0	0	25,0000	25,0000	0,0000	$1.750,00	5000,00	0,00
UK	12	2026-07-06	13,2813	0	0	13,2813	13,2813	0,0000	$929,69	2500,00	500,00
UK	13	2026-07-13	13,2813	0	0	13,2813	13,2813	0,0000	$929,69	2500,00	500,00
UK	14	2026-07-20	13,2813	0	0	13,2813	13,2813	0,0000	$929,69	2500,00	500,00
UK	15	2026-07-27	5,0000	0	0	5,0000	5,0000	0,0000	$350,00	1000,00	0,00

Alterrnative Books partially empty 20fts, paying a useless "speed premium" for inventory that will just sit in warehouses.

Market	Week	Week_Date	Total_Demand_CBM	n40	n20	LCL_CBM	Coverage_CBM	Slack_CBM	Cost	Units_C	Units_D
UK	4	2026-05-11	24,9900	0	1	0,0000	28,0000	3,0100	$2.500,00	4998,00	0,00
UK	5	2026-05-18	20,8300	0	1	0,0000	28,0000	7,1700	$2.500,00	4165,00	0,00
UK	6	2026-05-25	25,0000	0	1	0,0000	28,0000	3,0000	$2.500,00	5000,00	0,00
UK	7	2026-06-01	25,0000	0	1	0,0000	28,0000	3,0000	$2.500,00	5000,00	0,00
UK	8	2026-06-08	25,0000	0	1	0,0000	28,0000	3,0000	$2.500,00	5000,00	0,00
UK	9	2026-06-15	25,0000	0	1	0,0000	28,0000	3,0000	$2.500,00	5000,00	0,00
UK	10	2026-06-22	25,0000	0	1	0,0000	28,0000	3,0000	$2.500,00	5000,00	0,00
UK	11	2026-06-29	25,0000	0	1	0,0000	28,0000	3,0000	$2.500,00	5000,00	0,00
UK	12	2026-07-06	13,2813	0	0	13,2813	13,2813	0,0000	$929,69	2500,00	500,00
UK	13	2026-07-13	13,2813	0	0	13,2813	13,2813	0,0000	$929,69	2500,00	500,00
UK	14	2026-07-20	13,2813	0	0	13,2813	13,2813	0,0000	$929,69	2500,00	500,00
UK	15	2026-07-27	5,0000	0	0	5,0000	5,0000	0,0000	$350,00	1000,00	0,00

Shipping mode to Australia

Locks at 100% LCL since weekly demand (1-3 CBM) is far below FCL break-even, ensuring zero slack.

Market	Week	Week_Date	Total_Demand_CBM	n40	n20	LCL_CBM	Coverage_CBM	Slack_CBM	Cost	Units_E
AU	4	2026-05-11	0,0000	0	0	0,0000	0,0000	0,0000	$0,00	0,00
AU	5	2026-05-18	0,0000	0	0	0,0000	0,0000	0,0000	$0,00	0,00
AU	6	2026-05-25	0,0000	0	0	0,0000	0,0000	0,0000	$0,00	0,00
AU	7	2026-06-01	0,9375	0	0	0,9375	0,9375	0,0000	$32,81	300,00
AU	8	2026-06-08	0,0000	0	0	0,0000	0,0000	0,0000	$0,00	0,00
AU	9	2026-06-15	0,0000	0	0	0,0000	0,0000	0,0000	$0,00	0,00
AU	10	2026-06-22	0,9375	0	0	0,9375	0,9375	0,0000	$32,81	300,00
AU	11	2026-06-29	0,0000	0	0	0,0000	0,0000	0,0000	$0,00	0,00
AU	12	2026-07-06	2,1875	0	0	2,1875	2,1875	0,0000	$76,56	700,00
AU	13	2026-07-13	0,9375	0	0	0,9375	0,9375	0,0000	$32,81	300,00
AU	14	2026-07-20	0,9375	0	0	0,9375	0,9375	0,0000	$32,81	300,00
AU	15	2026-07-27	0,9375	0	0	0,9375	0,9375	0,0000	$32,81	300,00

→ Maintain 100% LCL to preserve perfect agility and cost control in an ultra-lean market.

A2. Shipping mode decision & Cost justification

Summary of all markets

Baseline

Market	Weeks	Total_Demand_CBM	Total_n40	Total_n20	Total_LCL_CBM	Total_Coverage_CBM	Total_Slack_CBM	Total_Cost
AU	4 --> 15	6,8750	0	0	6,8750	6,8750	0,0000	$240,63
UK	4 --> 15	240,6638	0	0	240,6638	240,6638	0,0000	**$16.846,46**
US	4 --> 15	379,2169	2	10	46,3209	456,3209	77,1041	**$49.664,19**

- **FCL & LCL Usage**: Books fewer FCLs (12 containers) and utilizes higher LCL volume (294 CBM).
- **Coverage & Slack**: Achieves 100% Coverage while keeping Slack to an absolute minimum (77 CBM).
- **Final Cost**: $66 751
- Why it works: The algorithm uses LCL strictly as a volume buffer. FCLs are only booked when they reach 100% utilization. It perfectly balances capacity and cost.

Alterrnative

Market	Weeks	Total_Demand_CBM	Total_n40	Total_n20	Total_LCL_CBM	Total_Coverage_CBM	Total_Slack_CBM	Total_Cost
AU	4 --> 15	6,8750	0	0	6,8750	6,8750	0,0000	$240,63
UK	4 --> 15	240,6638	0	8	44,8438	268,8438	28,1800	**$23.139,06**
US	4 --> 15	379,2169	6	6	9,0234	567,0234	187,8066	**$51.004,69**

- **FCL & LCL Usage**: Aggressively overbooks FCLs (20 containers) while minimizing LCL usage (61 CBM).
- **Coverage & Slack**: Achieves 100% Coverage, but generates a massive 216 CBM of Slack (+280% wasted space).
- **Final Cost**: $74 384
- Why it happens: Human planners force volume into partially empty FCLs just to avoid the 5-day LCL delay. This creates an expensive "speed premium" where we literally pay to ship empty air.

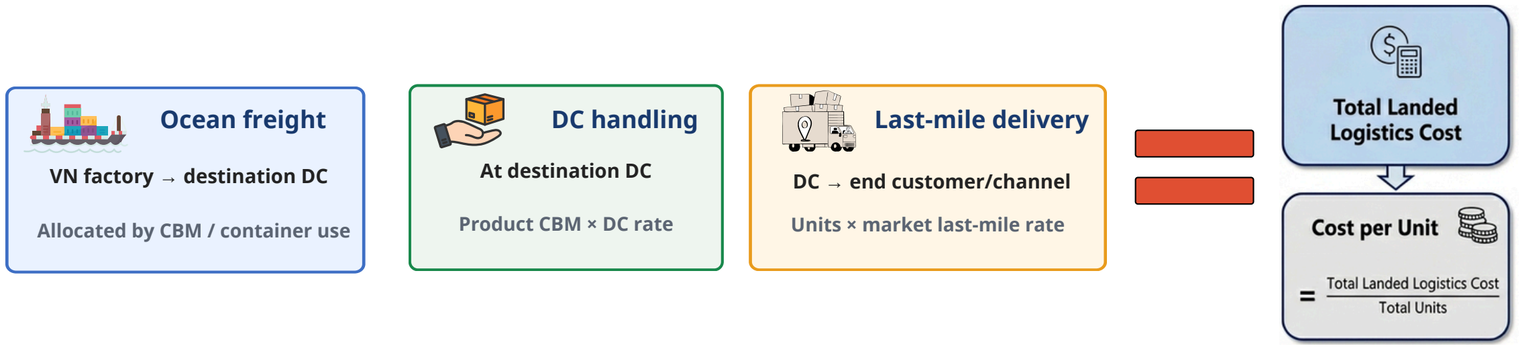
Why this is the right trade-off?

Timing Constraint	We intentionally traded a 5-day LCL consolidation delay for financial efficiency. Paying a premium for "speed" during an off-peak season holds no strategic value and only inflates warehouse holding costs.
Volume Constraint	Instead of forcing residual volume into FCLs (which caused a 280% increase in Slack), we used LCL as a flexible buffer. This guarantees 100% space utilization for every booked container.
Cost Constraint	By refusing to pay a $7633 overcharge to ship "empty air", our hybrid approach successfully balances capacity and transit time to secure the absolute minimum Total Landed Cost.

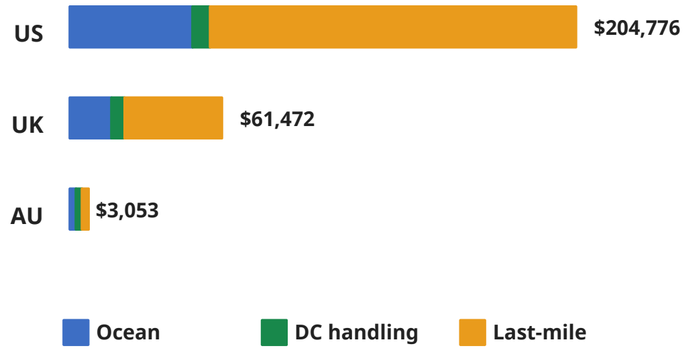
A3. Total Landed Cost

A3 calculates market-level landed logistics cost from three transparent components: ocean freight from Vietnam to DC, DC handling cost, and last-mile delivery from DC to end customer.

Cost architecture used in calculating total landed logistics cost



Total cost by component (USD)



Total landed cost summary by market

Market	Units	CBM	Ocean freight	DC handling	Last-mile	Total landed cost	Cost/unit
US	184,992	379.22	$49,664.19	$7,117.78	$147,993.60	**$204,775.57**	**$1.11**
UK	49,164	240.66	$16,846.46	$5,294.60	$39,331.20	**$61,472.27**	**$1.25**
AU	2,200	6.88	$240.63	$171.88	$2,640.00	**$3,052.50**	**$1.39**

Key Cost Insights

- **US** absorbs the **largest total landed logistics cost** at **$204,775.57 (184,992 units)**, accounting for **~76% of total spend**; however, it has the **lowest cost per unit at $1.11**, indicating the **best scale efficiency**.
- **UK** is the **mid-sized market with $61,472.27 total landed cost and $1.25 per unit;** its **cost structure is more balanced than AU,** but last-mile still remains the main cost driver.
- **AU** has the **smallest total spend at $3,052.50**, yet the **highest cost per unit at $1.39**, showing that **low volume reduces scale efficiency and makes the market more expensive to serve on a per-unit basis.**
- Across all three markets, **last-mile delivery is the dominant cost layer (US: 72.3%, UK: 64.0%, AU: 86.5%** of total landed cost), meaning the **biggest optimization opportunity lies in downstream delivery design, not ocean freight alone**.

Excel reference: LANDED_COST for row-level formulas; COST_OVERVIEW for market summary and A2 reconciliation.

A3. Total landed cost

Absolute logistics spend is highest in the US, but the true efficiency risk appears in Australia where logistics cost takes the largest share of product value

Market landed cost and burden ranking						
Market	**Total cost**	**Cost/unit**	**Avg ex-works price**	**Burden**	**Rank**	**Decision read**
US	$204,775.57	$1.11	$92.93	**1.19%**	#3	Largest spend, but lowest burden
UK	$61,472.27	$1.25	$78.78	**1.59%**	#2	Mid-level burden, CBM-heavy
AU	$3,052.50	**$1.39**	$30.00	**4.63%**	**#1**	**Highest relative logistics burden**

Logistics burden %



Key interpretation

- **US** is the **largest cost pool** but the **most efficient market on a unit-cost basis**. Its total **landed logistics cost is the highest** because it carries the **largest volume**, but **the burden remains low at 1.19% due to higher average ex-works price and better scale absorption**.
- **UK** shows a **moderate burden profile**. **Its logistics burden of 1.59%** suggests that costs are still manageable, although CBM-heavy volume makes it **less efficient than the US.**
- **AU** is the **real cost-efficiency hotspot**. Despite the **smallest total spend**, AU has the **highest cost/unit and the lowest average ex-works price**, pushing logistics burden up to **4.63%**.

Strategic Implication for MK

Highest burden market: Australia - 4.63%

Why it matters:

AU's logistics cost is small in absolute dollars, but it consumes the largest share of product value. This means margin risk is higher in AU than what total cost alone suggests.

Decision implication:

Prioritize AU for last-mile cost control, shipment consolidation, and distributor-level delivery efficiency. Do not force **FCL** purely for ocean cost saving if it increases downstream inefficiency or creates low-volume cost absorption issues.

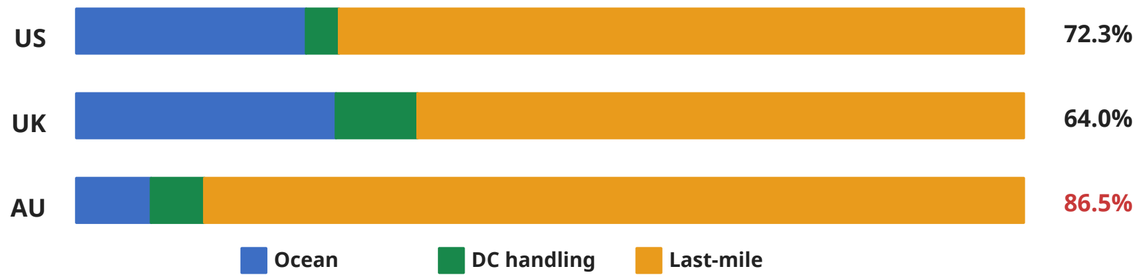
Bottom line: The US is the largest spending market, but Australia is the highest burden market; therefore, AU should be managed for margin protection, not just shipment cost reduction

Excel reference: COST_OVERVIEW rows 4–6 for total landed logistics cost, weighted average ex-works price, and burden ranking.

A3. Total landed cost

Cost-driver analysis shows that AU's burden is driven by last-mile delivery after DC arrival. The improvement lever should therefore focus on last-mile cost control rather than FCL switching

Logistics Cost Structure by Market



Breakdown of Logistics Cost Components

Market	Ocean	DC handling	Last-mile	Last-mile share
US	$49,664	$7,118	$147,994	72.3%
UK	$16,846	$5,295	$39,331	64.0%
AU	$241	$172	**$2,640**	**86.5%**

Cost burden diagnosis

- **AU** has the **smallest shipment scale (2,200 units / 6.88 CBM)**, so **ocean freight** is only **$241** and not the **main burden driver**.
- **Last-mile delivery** contributes **$2,640**, accounting for **86.5% of AU landed logistics cost**.
- Because **AU volume** is far below **FCL threshold, switching to FCL** would increase **empty-space cost rather than solve the burden issue**.
- The practical lever is to **reduce the last-mile delivery cost,** while **keeping AU on LCL for flexibility.**

Recommended action and expected impact

Action: Keep AU on LCL due to low shipment volume, but reduce downstream delivery cost by bundling distributor replenishment cycles and renegotiating AU last-mile rates.

Illustrative impact: A 10% reduction in AU last-mile rate ($1.20 → $1.08/unit) saves $264 over the planning period and lowers AU logistics burden from 4.63% to 4.23%

Bottom line: AU is the highest-burden market, but the root cause is last-mile intensity, not ocean freight. MK should keep AU on LCL and improve cost-to-serve through downstream delivery consolidation and rate negotiation

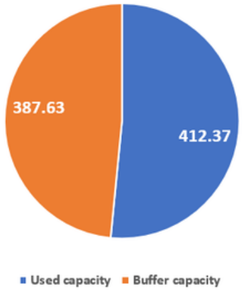
Excel reference: COST_OVERVIEW for cost component split; LANDED_COST for ocean/DC/last-mile row-level formulas.

B. DISTRIBUTOR RECOMMENDATION

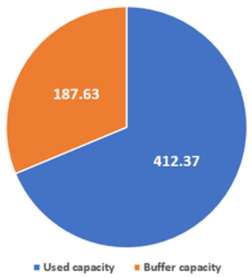
1. DISTRIBUTOR SELECTION - USA MARKET

1. Can they handle the volume?

PLAYGIANT



KIDWORLD DISTRIB



Both have **sufficient warehouse capacity**, however, **Playgiant** provides a **stronger capacity buffer**

2. Do they reach the right customers?

The U.S. market is **online-first**, with Amazon and Walmart.com accounting for **62%**. Consumer demand is **seasonal**, requiring **fast (2-day) delivery.**

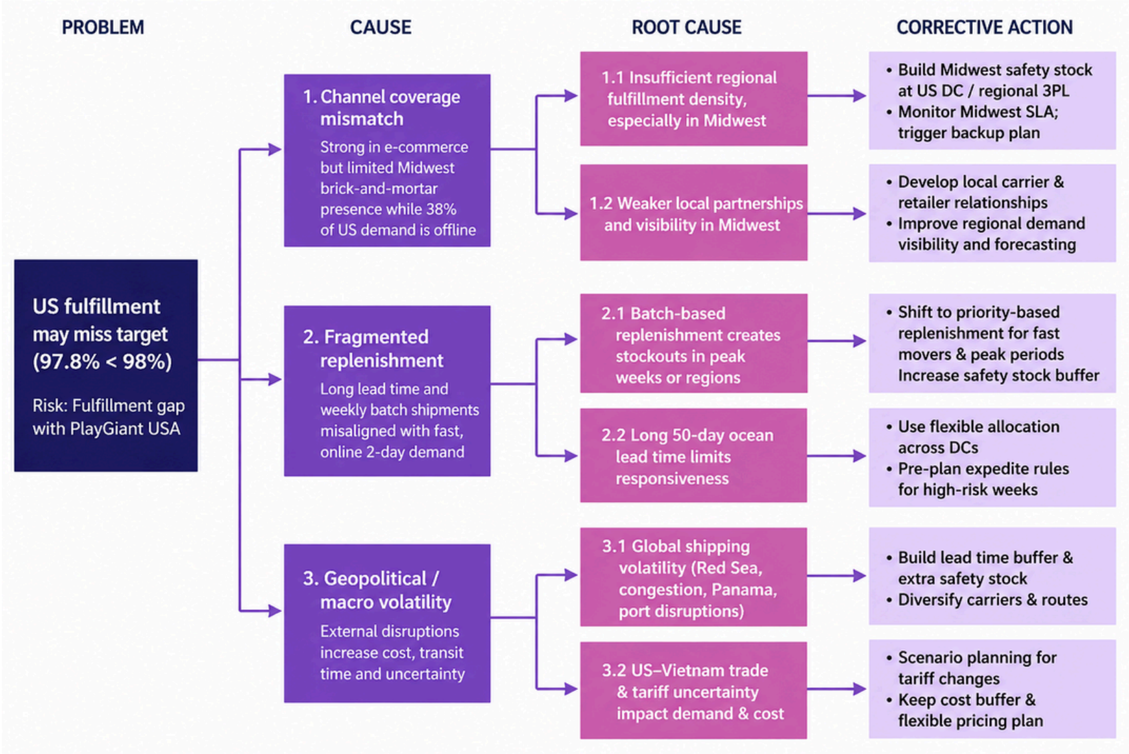
Distributor	Capacity (CBM)	Fulfillment Rate	Key Strength / Weakness
PlayGiant USA	800	978%	Strong in e-commerce; limited Midwest brick-and-
KidWorld Distrib.	600	963%	Strong Q4 peak performance; good shelf

Playgiant fits US customers behavior and expectations **better.**

Playgiant should be selected as primary distributor in US

3.What is the key risk?

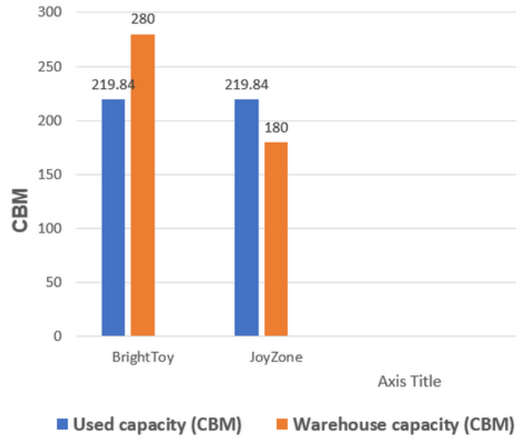
Key risk: Fulfillment gap (97.8%<98%)



B. DISTRIBUTOR RECOMMENDATION

2. DISTRIBUTOR SELECTION - UK MARKET

1. Can they handle the volume?



BrightToy can **handle** the UK volume with **60.16 CBM spare capacity**, while **JoyZone** is **over capacity** by 39.84 CBM.

BrightToy wins

2. Do they reach the right customers?

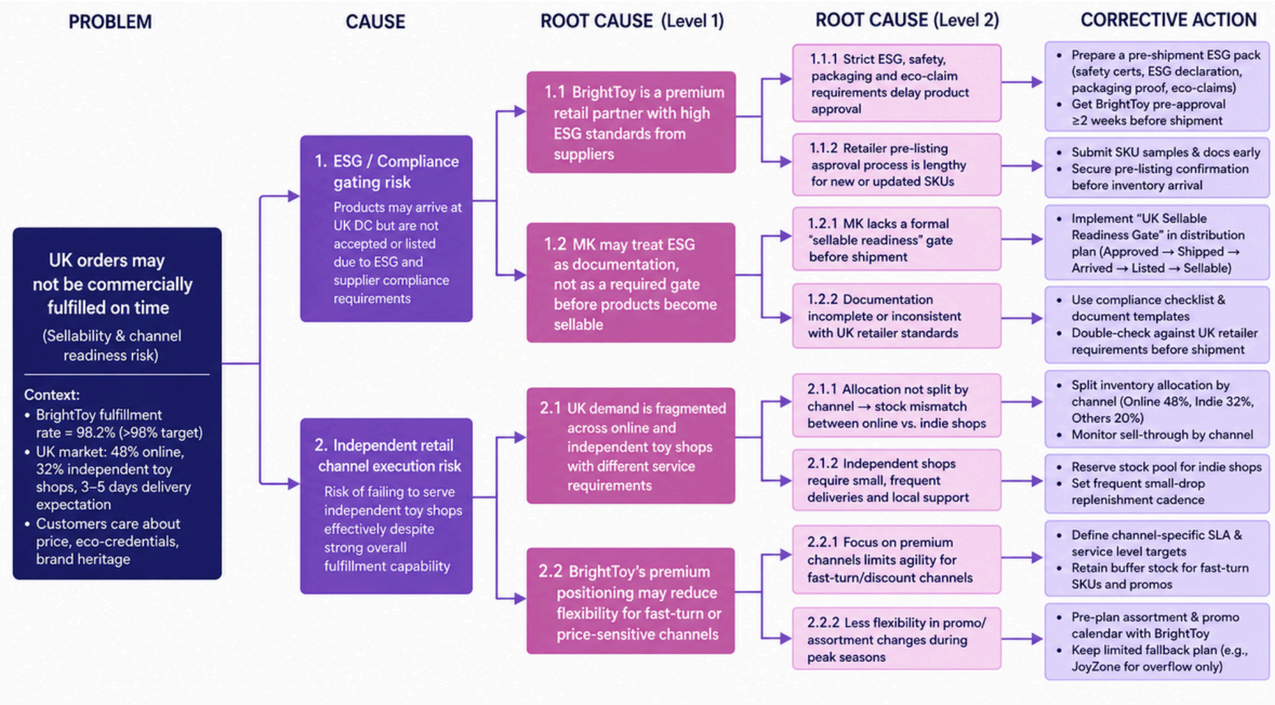


BrightToy which is premium retail partner with **high ESG standards** and **meets required fulfillment rate** (98.2%>98%) fits the UK market better than **JoyZone whose only 91.5%**.

BrightToy should be chosen as main distributor in UK

3.What is the key risk?

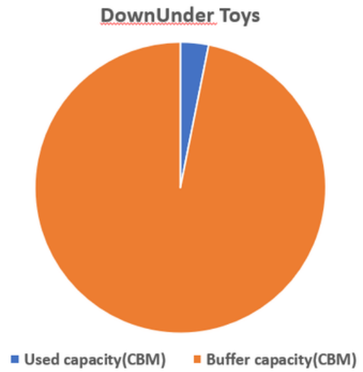
Key risks : Late delivery risks



B. DISTRIBUTOR RECOMMENDATION

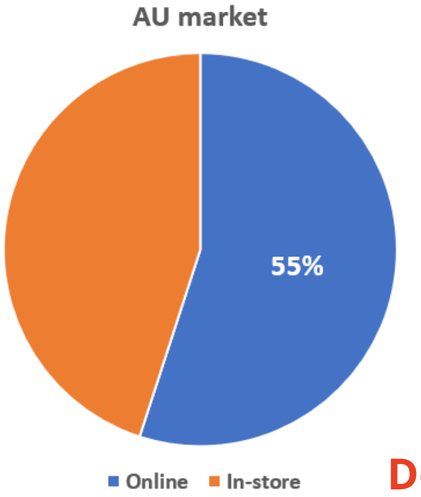
3. DISTRIBUTOR SELECTION - UK MARKET

1. Can they handle the volume?



DownUnder Toys easily meets AU needs: 220 CBM capacity vs. 6.875 CBM planned volume.

2. Do they reach the right customers?



Australia has a **balanced market**, **3–7 day delivery** expectations, and **safety-driven purchases.**

DownUnder Toys has **strong NSW and VIC coverage** (major demand centers) .

DownUnder Toys is the only viable

3.What is the key risk?

Key risk: Single-point-of-failure &regional coverage risk

PROBLEM	CAUSE	ROOT CAUSE (LEVEL 1)	ROOT CAUSE (LEVEL 2)	CORRECTIVE ACTION
AU orders may not be fulfilled reliably (Single-point-of-failure & regional coverage risk) Context: • DownUnder Toys is the only national AU distributor in the shortlist • Fulfillment rate = 96.8% (< 98% target) • Strength concentrated in NSW & VIC • AU market: 55% online, 45% in-store, 3-7 day delivery expectation	1. Single distributor dependency MK relies on one below-target distributor with no clear fallback if service drops	1.1 No alternative AU distributor in the shortlist	1.1.1 MK has no pre-qualified backup partner, 3PL, or overflow plan for AU	• Build an AU contingency layer • Pre-arrange emergency 3PL / overflow warehouse • Set trigger to activate backup support
		1.2 DownUnder's fulfillment rate is already below target	1.2.1 Low service buffer makes peak-season disruption harder to absorb	• Monitor SLA weekly • Escalate if fulfillment < 98% • Activate backup if rolling 4-week fulfillment < 95%
	2. Regional coverage concentration Service capability appears strongest in NSW & VIC, creating uneven nationwide execution risk	2.1 Coverage strength is concentrated in NSW & VIC	2.1.1 A national distributor does not guarantee equal service quality across all states	• Track SLA by state • Add regional delivery KPIs • Trigger support for underperforming states
		2.2 AU demand is mixed across online and in-store channels	2.2.1 Inventory may exist nationally but not be positioned close enough to regional demand	• Pre-position regional safety stock • Secure backup carriers for remote states • Allocate stock by state-level sell-through

B. DISTRIBUTOR RECOMMENDATION

4. CONTINGENCY: WHAT IF THEY FAIL?

1.Problem Definition: If distributor fulfillment drops to 88% for 4 consecutive weeks, creating a 10% fulfillment gap, MK would face **a service reliability risk** . The key issue is not warehouse capacity, but **fulfillment performance**: order processing, labor availability, channel fit, and operational reliability.

Market	Peak Season	Period of Demand	Lead Time	Shipment Request Date	4 Peak-Week Demand	Fulfilled at 88%	Unfulfilled at 88%	Target Unfulfilled at 98%	Excess Service Gap
US	August	1/8/2026 – 31/8/2026	50 days	12/6/2026 – 12/7/2026	**57,298 units**	**50,422 units**	**6,876 units**	**1,146 units**	**5,730 units**
UK	Nov–Dec	1/11/2026 – 31/12/2026	40 days	22/9/2026 – 21/11/2026	**10,050 units**	**8,844 units**	**1,206 units**	**201 units**	**1,005 units**
AU	Nov–Dec	1/11/2026 – 31/12/2026	25 days	7/10/2026 – 6/12/2026	**600 units**	**528 units**	**72 units**	**12 units**	**60 units**

2.Risk matrix

Market	Risk	Likelihood score	Consequence score	Risk score
US	US fulfillment may miss ML target (97.8% < 98%)	4	4	16 (High)
UK	UK orders may not be commercially fulfilled on time	3	4	12 (Moderate)
AU	AU orders may not be fulfilled reliably	4	5	20 (Extremely High)

3.Fallback option for AU

When no viable distributor alternative is available, MK should shift from "service recovery" to customer protection mode.

- **Prioritize critical orders:** allocate inventory first to top retailers, key accounts, best-selling SKUs, and high-demand regions.
- **Control demand pressure:** pause promotions, limit order quantities, and manage retailer quotas to prevent overselling.
- **Protect delivery trust:** use backorder promises, extend delivery windows, and communicate delays proactively.
- **Add emergency logistics support:** use temporary 3PL, premium freight, or direct DC-to-retailer shipment where possible.
- **Offer compensation:** provide free shipping, vouchers, discounts, or priority fulfillment for delayed customers.

- ***Financial cost:*** emergency 3PL, selective premium freight, direct DC-to-retailer shipment, temporary storage near NSW/VIC, lost sales from paused promotions, reasonable customer compensation.
- ***Operational cost:*** regional stock allocation, NSW/VIC prioritization, retailer order limits, daily SKU-level inventory tracking, backorder control, proactive retailer/customer communication.

B. DISTRIBUTOR RECOMMENDATION

4. CONTINGENCY: WHAT IF THEY FAIL?

4.Fallback option for US market:

MK should not fully switch. MK should use a split-switch model.

Assumption: MK reallocates US peak-season demand between PlayGiant and KidWorld to achieve a minimum **blended fulfillment rate of 90%.**

Formula: m1 (volume for Playgiant) +m2 (volume for Kidworld) = 562,625

88% × m1 + 96.3% × m2 = 90% × 562,625

Distributor	Allocation	Share of Total Volume	Fulfillment Rate	Expected Fulfilled Units
PlayGiant	43,491 units	759%	88%	38,272 units
KidWorld	13,807 units	241%	963%	13,296 units
Total	57,298 units	100%	90% blended	51,568 units

Units rescued = Fulfilled units after fallback − Fulfilled units without fallback= 51,568 − 50,422 = 1,146 units

- ***Financial switching cost***: KidWorld onboarding, EDI/system integration, inventory re-routing, rush freight, dual-distributor management cost.
- ***Operational switching cost:*** split inventory, channel-based order allocation, retail/overflow shift to KidWorld, stock imbalance risk, daily fulfillment monitoring.

5.Fallback option for UK market:

MK should use JoyZone as a selective overflow fallback, not as a full replacement.

Assumption: MK reallocates US peak-season demand between PlayGiant and KidWorld to achieve a minimum **blended fulfillment rate of >90%.**

Formula: m1 (volume for BrightToy) +m2 (volume for JoyZone) = 59,040

88% × m1 + 91.5% × m2 = 90% × 59,040

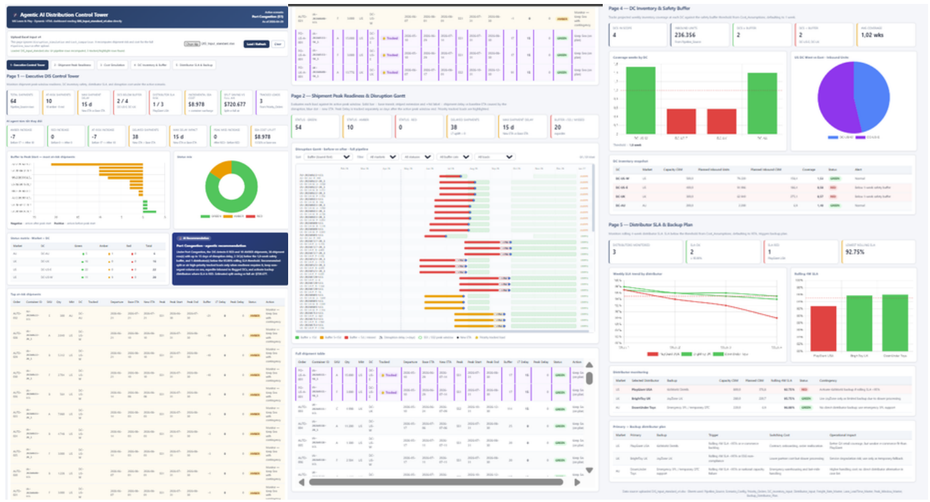
Distributor	Allocation	Share of Total Volume	Fulfillment Rate	Expected Fulfilled Units
BrightToy	0 units	0%	88%	0 units
JoyZone	10,050 units	100%	915%	9,196 units
Total	10,050 units	100%	91.5% blended	9,196 units

Units rescued = Fulfilled units after fallback − Fulfilled units without fallback = 9,196 − 8,844 = 352 units

- ***Financial switching cost:*** inventory re-routing, retailer compensation, lost premium sales, shelf-space risk, margin pressure.
- ***Operational switching cost:*** channel mismatch, priority order segmentation, slower JoyZone turnaround, top-retailer protection, separate retailer communication.

C1. Decision Logic

The Distribution Intelligence System (DIS) is an agentic control tower that continuously monitors shipment ETA, DC inventory, distributor SLA, and freight cost, then detects exceptions, simulates recovery options, recommends actions, and reports alerts to management with AI agent teams.



The HTML dashboard has 5 pages: Executive Control Tower, Shipment Peak Readiness, Cost Simulation, DC Inventory & Buffer, and Distributor SLA & Backup.

Link to Dashboard

Access the dashboard link, upload the **input_standard.xlsx** file using the “Choose File” button, then click “Load / Refresh” at the top of the dashboard. **Detailed usage instructions are provided in the Appendix A.**

AI Agent Workflow

Agent	Input	Automated task	Output	Dashboard location
Monitoring Agent	Shipment ETA, DC inventory, SLA	Reads refreshed Excel data, checks active scenario, tracks KPI changes	Abnormal signal list	Page 1, 2, 4, 5
Risk Detection Agent	New ETA, deadline, peak window, safety buffer	Classifies shipment/DC/distributor risk	RED / AMBER / GREEN	Page 1, 2, 4, 5
Cost Simulation Agent	Sea cost, air cost, qty, weight	Compares keep-sea, full-air, split options	Cost comparison table	Page 3
Recommendation Agent	Risk + cost + priority	Selects best response option	Sea / Air / Split / Backup	Page 1, 3, 5
Reporting Agent	Final decision output	Converts analysis into management summary	Dashboard alert / executive message	Page 1

→ This agent is presented as an automation module within DIS, not a "fully autonomous AI." The planner still approves the final decision.

Decision Rules Used by the Agentic DIS

Rule	IF	THEN	Agent Owner
R1	Port congestion S1 is active	Activate disruption model	Monitoring Agent
R2	DC = DC-US-E or DC-UK	Add +15 days LT and +20% sea cost	Monitoring + Cost Agent
R3	New ETA > Base ETA	Mark shipment as delayed	Monitoring Agent
R4	New ETA ≤ peak start	Classify GREEN	Risk Detection Agent
R5	New ETA within peak window	Classify AMBER	Risk Detection Agent
R6	New ETA > peak window end	Classify RED	Risk Detection Agent
R7	High-priority order delayed vs required date	Trigger recovery simulation	Cost Simulation Agent
R8	Full air is too costly but service risk exists	Recommend split shipment	Recommendation Agent
R9	DC coverage <1 week OR SLA <95%	Trigger buffer/SLA alert	Risk Detection Agent
R10	SLA RED or DC RED persists	Activate backup / expedite action	Recommendation + Reporting Agent

→ The DIS converts disruption response from manual judgment into a repeatable rule-based decision process

C2. Disruption Scenario

Disruption sceanrio setup

Affected lanes	Lead time increase	Container cost increase	Airfreight rate	Product A weight	DC safety buffer	Distributor SLA threshold
DC-US-E and DC-UK	+15 days	20%	USD 4.50/kg	1.2 kg/unit	1 week average demand	95% rolling 4 weeks

Disruption Impact: ETA Delay & Readiness

Under S1 Port Congestion, DIS detects ETA delays on affected US-E and UK lanes. While the overall situation is still manageable with 38 delayed shipments and a maximum delay of 15 days, a critical impact is observed with 7 shipments missing their original schedules (SS1) and being pushed to SS2. Despite this, shipment readiness remains stable with 10 AMBER / 0 RED, indicating that peak-window targets are still being met.

KPI after S1	Result	Meaning
Total shipment rows	64	Full pipeline monitored
Delayed shipments	38	New ETA > Base ETA
Max shipment delay	15 days	Direct effect of port congestion
Shipment readiness	10 AMBER / 0 RED	Timing buffer reduced, but no peak-window miss
Missed Schedule	7 shipments	Change to SS2 (Critical delay)

Service Risk & Agentic Response

Even with 0 shipment RED, service risk still exists downstream: 2 DCs below safety buffer and 1 distributor below SLA threshold. DIS therefore triggers mitigation, not only monitoring.

Risk area	Result	Agentic response
DC buffer risk	2 / 4 DCs below buffer	Expedite / rebalance DC-US-E and DC-UK
Distributor SLA risk	1 / 3 at risk	Prepare KidWorld backup for PlayGiant USA
Agent workflow	Monitor → Detect → Recommend → Report	Turn KPI alerts into planner-approved

Cost Comparison: Three Recovery Options

The cost simulation shows that full airfreight is financially excessive, while keeping all volume on sea is cheapest but does not protect urgent demand. The 40/60 split option gives a better balance between service recovery and cost control.

Option	Total cost	Cost implication	Service implication	Decision
Keep sea after disruption	**$75,180**	Lowest cost; only +$8,978 vs base sea	Slowest recovery; urgent volume remains exposed	Use for non-urgent volume
Full airfreight	**$1,276,322**	Highest cost; +$1.20M vs disrupted sea	Fastest recovery	Avoid unless absolutely critical
Split shipment 40/60	**$555,646**	Saves **$720,677 vs full air**	Protects urgent volume while keeping rest on sea	Recommended for tracked/high-risk loads

Cost Interpretation & Final Choice

Decision implication: The best decision is not the cheapest option, but the option that protects service at a controlled premium. Therefore, DIS recommends:

- Keep sea for standard / non-urgent shipments.
- Use split shipment for tracked or high-risk priority loads.
- Reserve full airfreight only for extreme cases (split cannot meet required date).

Comparison	Result	Meaning
Base sea → Disrupted sea	$66,203 → $75,180	Port congestion adds **$8,978**, or around **13.56%** uplift
Disrupted sea → Full air	$75,180 → $1,276,322	Full air adds over **$1.20M**, too expensive for full-network recovery
Full air → Split	$1,276,322 → $555,646	Split reduces premium by **$720,677**
Disrupted sea → Split	$75,180 → $555,646	Split costs more than sea, but buys service protection for urgent demand

C3. Monitoring Dashboard

Dashboard Page 1 & 2: Monitoring Shipment Exceptions in Real Time



Page 1: Executive Control Tower shows total shipments, at-risk shipments, max delay, DC buffer risk, SLA risk, incremental sea cost, split saving.



Page 2: Gantt shows base ETA, disruption extension, new ETA, and priority tracked loads.

Agent roles

Dashboard component	Agent function
KPI cards	Reporting Agent summarizes network status
Status matrix	Risk Detection Agent aggregates RED/AMBER/GREEN by market/DC
Gantt chart	Monitoring Agent tracks ETA movement
Tracked priority loads	Recommendation Agent prioritizes recovery

C3. Monitoring Dashboard

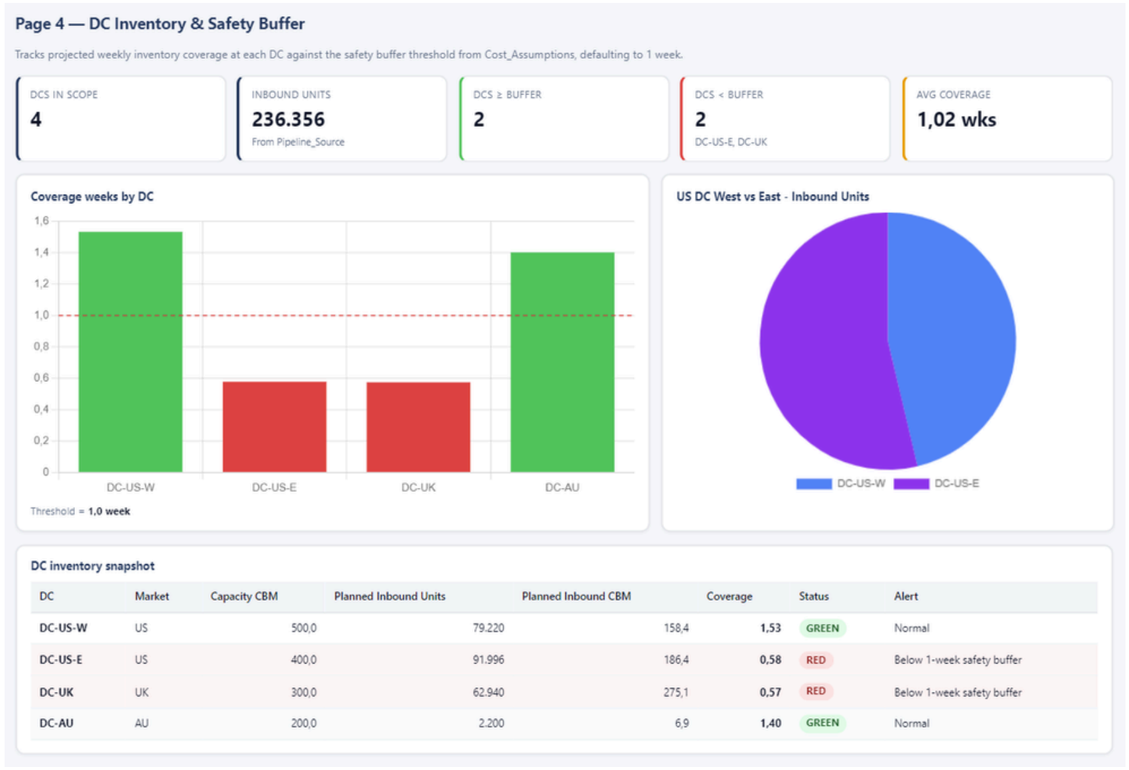
Dashboard Page 4 & 5: Monitoring Inventory Buffer and Last-Mile Service Risk

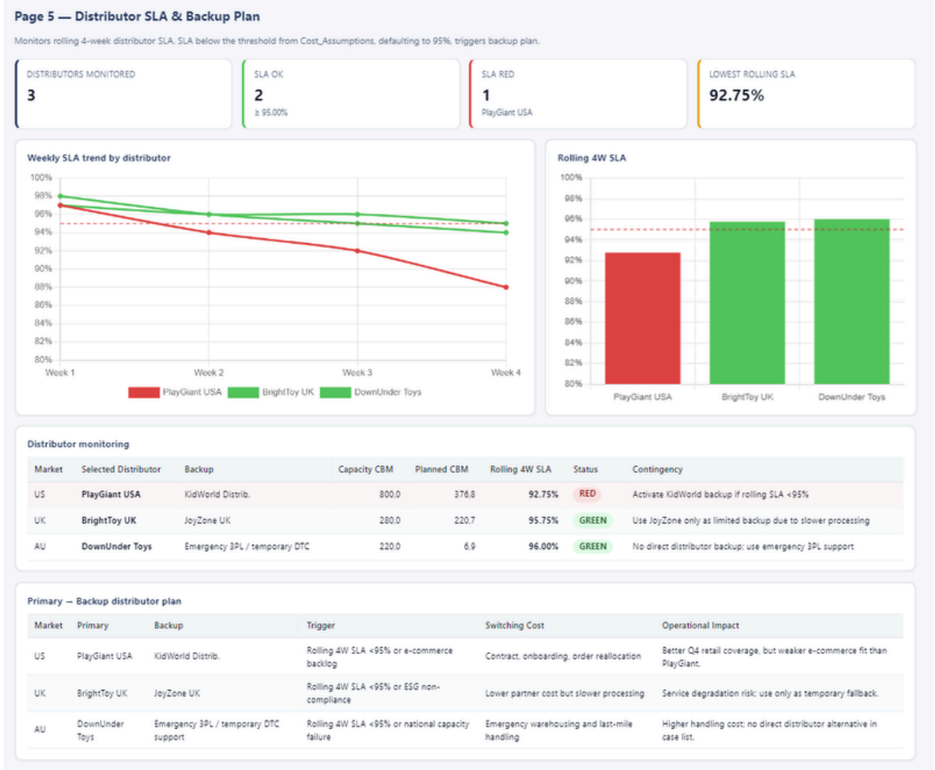
Agent roles

DC-US-E below 1-week buffer	Agent action
DC-US-E below 1-week buffer	Risk Detection Agent raises RED alert
DC-UK below 1-week buffer	Recommendation Agent suggests expedite
PlayGiant USA SLA = 92.75%	Recommendation Agent activates KidWorld
Management update	Reporting Agent writes dashboard alert

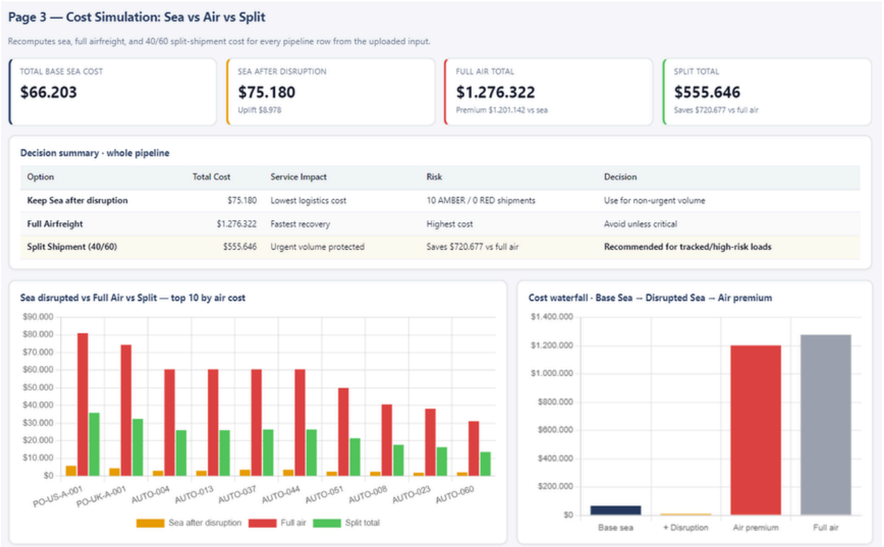
Current output

KPI	Result
DCs below buffer	2/4/2026
DC-US-E coverage	0.58 week
DC-UK coverage	0.57 week
SLA RED	1 distributor
Lowest rolling SLA	92.75% — PlayGiant USA





C3. Monitoring Dashboard: Page 3



Option	Total cost	Cost implication	Recommended use
Keep sea after disruption	**$75,180**	Lowest cost, but weak for urgent service recovery	Use for standard / non-urgent volume
Full airfreight	**$1,276,322**	Fastest but financially excessive for full network	Reserve only for extreme critical cases
Split shipment 40/60	**$555,646**	Saves **$720,677 vs full air** while protecting urgent volume	Recommended for tracked / high-risk loads

Page 3 turns disruption response into a cost-service trade-off: keep sea for non-urgent volume, use split shipment for tracked/high-risk loads, and reserve full airfreight only for extreme cases.

4. AI Values



AI creates the highest value in MK's DIS by **moving disruption management** from **manual reaction** to **early exception detection and optimized recovery.** The system uses **shipment ETA, lane disruption signals, DC inventory coverage, distributor SLA, priority order data, and freight rates** to **predict** which **orders** are most likely to **miss service targets**. It then **compares** recovery options such as **keeping sea freight, using full airfreight, splitting urgent** and **non-urgent volume**, or **activating backup distributors**. This improves customer fulfillment, reduces emergency logistics cost, shortens response time, and gives management a clear view of where intervention is required.

Thank You

APPENDIX A

HOW TO USE DASBOARD AND CHANGE DATA INPUT?

Step 1 — Prepare the input file Link to sheet

Use the original **input_standard.xlsx** file. If no data needs to be changed, upload this file directly.

Step 2 — Change scenario if needed

Open **Scenario_Config.** Set the selected scenario as **Active = TRUE** and all other scenarios as **Active = FALSE**. Only one scenario should be active.

Scenario ID	Scenario Name	Active	US-E LT Increase	UK LT Increase
S0	Baseline	☐	0	0
S1	Port Congestion	☑	15	15

Step 3 — Change monitored shipment if needed

Open **Priority_Orders**. Update the load/container ID and related priority order information. Make sure the ID matches the same ID in **Pipeline_Source**.

Order ID	Container ID	SKU	Qty	Market	DC	Priority
PO-US-A-001	US-20260511-40_1	A	20,000	US	DC-US-E	High
PO-UK-A-001	UK-20260608-40_1	A	13,773	UK	DC-UK	High

Step 4 — Save the file

Save the updated Excel file as .xlsx.

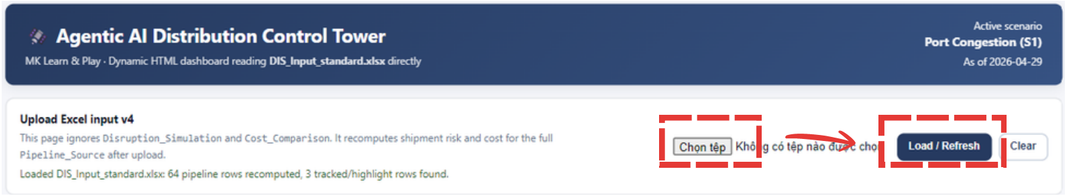
Step 5 — Open the dashboard

Open the dashboard link or the HTML dashboard file in a web browser. Link to Dashboard or use html file

Step 6 — Upload the Excel input

At the top of the dashboard, click **“Choose File”.**

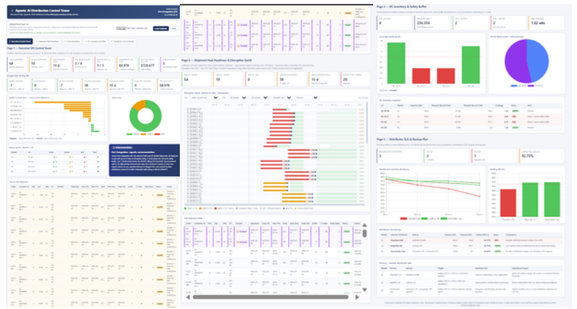
Select the updated **input_standard.xlsx** file, or the original file if no changes were made. → Click **“Load / Refresh”.**



Step 8 — Check the results

After refreshing, check that the active scenario shown on the dashboard is correct.

Then confirm whether the selected priority shipment is highlighted, shipment readiness is updated, cost simulation is recalculated, and DC/SLA alerts appear correctly.



APPENDIX B

Excel Workbook Structure and Calculation Logic Map

No.	Sheet Name	Role	Description	Used For
1	Pipeline Master Table	Main Output A.1	Tracks shipment/load pipeline from Vietnam to destination DC by week, market, SKU, quantity, arrival date, gap days, status and proposed	A.1
2	Note for A.1	Explanation	Explains assumptions, formula logic, week scope W04–W15, lead time, gap days and status/action rules.	A.1
3	Weekly_All_Markets	Input / A.2 Output	Aggregated weekly logistics plan across US, UK, AU; used to validate shipment scope and market-level demand.	A.2 / A.3
4	All_Load_Detail	Input / A.2 Output	Detailed load-level shipment data by market, week, mode/load, CBM and cost logic.	A.1 / A.3
5	Summary_Market	Input / A.2 Summary	Market-level logistics summary including total demand CBM, FCL/LCL split, coverage CBM, slack CBM and total freight cost.	A.2 / A.3 Audit
6	FGs & Log information	Master Data	Product logistics information such as ex-works price, carton information, CBM and related cost parameters.	A.3
7	LANDED_COST	Main Output A.3	Allocates ocean freight and calculates landed cost per SKU/load/market based on pipeline and logistics data.	A.3
8	COST_OVERVIEW	Main Output A.3	Summarizes landed cost, weighted ex-works price, logistics burden and reconciliation to A.2.	A.3
9	US	Supporting Input	US weekly shipment/load planning source.	A.1 / A.2
10	UK	Supporting Input	UK weekly shipment/load planning source.	A.1 / A.2
11	AU	Supporting Input	AU weekly shipment/load planning source.	A.1 / A.2

APPENDIX C

Contigency: 4 weeks demand in peak seasons for each market

Item name	Des	Market	Price ($)	Product group	Fixture	Weekly Capcity	Shipment Request	Demand Quantity
E	2026 Scooter ver	Australia	30	Outdoor & Sports Toy	789EF	3000	10/9/2026	300
E	2026 Scooter ver	Australia	30	Outdoor & Sports Toy	789EF	3000	11/13/2026	300
								600

Item name	Des	Market	Price ($)	Product group	Fixture	Weekly Capcity	Shipment Request	Demand Quantity
A	2026 Automatic Robot ver	US	105	Electronic Toy	123AB	15,000	6/19/2026	7,058
A	2026 Automatic Robot ver	US	105	Electronic Toy	123AB	15,000	6/26/2026	2,040
A	2026 Automatic Robot ver	US	105	Electronic Toy	123AB	15,000	7/3/2026	1,500
A	2026 Automatic Robot ver	US	105	Electronic Toy	123AB	15,000	7/3/2026	2,041
A	2026 Automatic Robot ver	US	105	Electronic Toy	123AB	15,000	7/10/2026	2,041
A	2026 Automatic Robot ver	US	105	Electronic Toy	123AB	15,000	7/10/2026	10,000
B	2025 Barbie ver colorful	US	55	Doll	123AB	15,000	6/19/2026	5,309
B	2025 Barbie ver colorful	US	55	Doll	123AB	15,000	6/26/2026	5,309
B	2025 Barbie ver colorful	US	55	Doll	123AB	15,000	7/3/2026	3,000
B	2025 Barbie ver colorful	US	55	Doll	123AB	15,000	7/10/2026	3,000
F	2025 Fantasy Spaceship Mo	US	90	Arts & Crafts Toy	789EF	3000	6/19/2026	4,500
F	2025 Fantasy Spaceship Mo	US	90	Arts & Crafts Toy	789EF	3000	6/26/2026	2,200
F	2025 Fantasy Spaceship Mo	US	90	Arts & Crafts Toy	789EF	3000	7/3/2026	5,000
F	2025 Fantasy Spaceship Mo	US	90	Arts & Crafts Toy	789EF	3000	7/10/2026	4,300
								57,298

Item name	Des	Market	Price ($)	Product group	Fixture	Weekly Capcity	Shipment Request	Demand Quantity
C	2026 Castle & Farm Blocks	UK	80	Building & Construc	456CD	5000	9/25/2026	750
C	2026 Castle & Farm Blocks	UK	80	Building & Construc	456CD	5000	10/2/2026	600
C	2026 Castle & Farm Blocks	UK	80	Building & Construc	456CD	5000	10/9/2026	600
C	2026 Castle & Farm Blocks	UK	80	Building & Construc	456CD	5000	10/16/2026	600
C	2026 Castle & Farm Blocks	UK	80	Building & Construc	456CD	5000	10/23/2026	600
C	2026 Castle & Farm Blocks	UK	80	Building & Construc	456CD	5000	10/30/2026	600
C	2026 Castle & Farm Blocks	UK	80	Building & Construc	456CD	5000	11/6/2026	500
C	2026 Castle & Farm Blocks	UK	80	Building & Construc	456CD	5000	11/13/2026	500
C	2026 Castle & Farm Blocks	UK	80	Building & Construc	456CD	5000	11/20/2026	500
D	2026 Animal Animation Book	UK	40	Educational Toy	456CD	5000	10/2/2026	360
D	2026 Animal Animation Book	UK	40	Educational Toy	456CD	5000	10/9/2026	360
D	2026 Animal Animation Book	UK	40	Educational Toy	456CD	5000	10/16/2026	360
D	2026 Animal Animation Book	UK	40	Educational Toy	456CD	5000	10/23/2026	360
D	2026 Animal Animation Book	UK	40	Educational Toy	456CD	5000	10/30/2026	360
D	2026 Animal Animation Book	UK	40	Educational Toy	456CD	5000	11/6/2026	1,000
D	2026 Animal Animation Book	UK	40	Educational Toy	456CD	5000	11/13/2026	1,000
D	2026 Animal Animation Book	UK	40	Educational Toy	456CD	5000	11/20/2026	1,000
								10,050